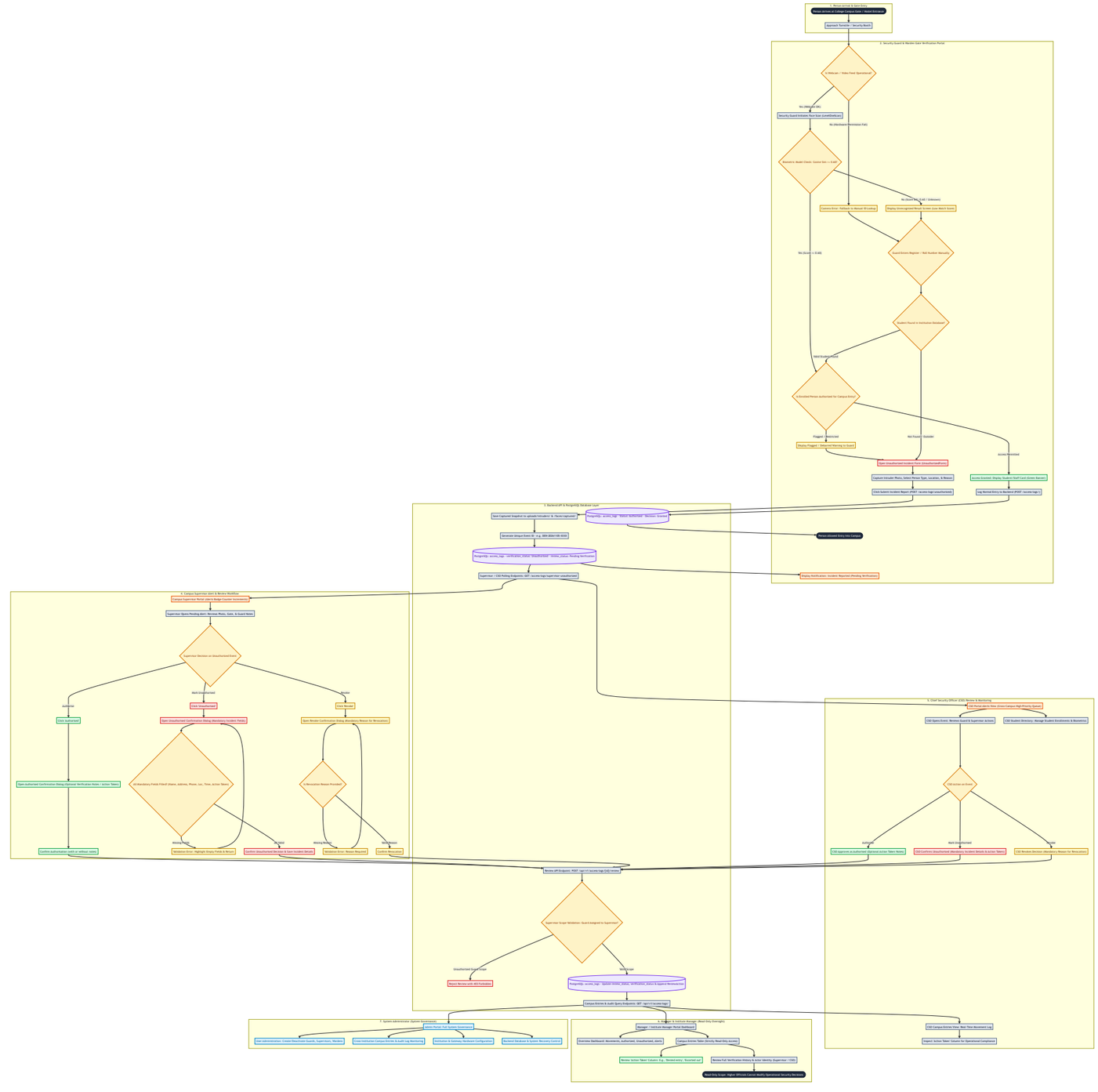
CAMPUS SECURITY APPLICATION — END-TO-END WORKFLOW FLOWCHART

Unified Architecture, Gate Facial Biometrics, Incident Adjudication, & Multi-Role Governance

PRODUCTION ARCHITECTURE 2026



Workflow Visual Legend & Role Conventions

- Start / End Boundary (Arrival / Granted)
- Decision Diamond (Threshold / Validation)
- Authorized Path (Granted / Permitted)
- Unauthorized Path (Denied / Incident)
- Pending Verification Alert Queue
- Exception / Fallback / Revocation
- PostgreSQL & API Persistence Layer
- System Administrator & Audit Control

Status Transition & Lifecycle Reference Matrix

Previous Status	Triggering Action	Acting Role	New Status	Verification Status	Action History Stored	Higher Authority Visibility
None (Arrival)	Face Scan Recognized & Authorized (Cosine Sim ≥ 0.60)	Security Guard / Warden	Completed	Authorized	Access Log created with normal entry timestamp	Visible in Campus Entries as Authorized
None (Arrival)	Unrecognized Face or Flagged Person (Fallback Unmatched)	Security Guard / Warden	Pending Verification	Unauthorized	Incident record with captured photo, gate ID, & guard notes	Alerts queue badge counter increments in real-time
Pending Verification	Authorise with optional notes / Action Taken	Campus Supervisor / CSO	Authorised	Authorized	ReviewAction with supervisor name, employee code, timestamp	Status updated; optional notes visible in entry details
Pending Verification	Mark Unauthorised (Mandatory Incident Details & Action Taken)	Campus Supervisor / CSO	Unauthorised	Unauthorized	ReviewAction with Full Name, Address, Phone, Action Taken	**Action Taken column displays recorded action**
Pending Verification	Revoke with mandatory justification	Campus Supervisor / CSO	Revoked	Revoked	ReviewAction with mandatory revocation reason	Marked Revoked; locked against further modifications
Authorised	Re-evaluate as Unauthorised	Chief Security Officer (CSO)	Unauthorised	Unauthorized	Appends second ReviewAction with updated incident fields	Real-time update across all higher-official portals
Unauthorised	Re-evaluate as Authorised	Chief Security Officer (CSO)	Authorised	Authorized	Appends second ReviewAction with authorization notes	Real-time update across all higher-official portals
Revoked	Subsequent Review Attempt	Any Role	**LOCKED**	**LOCKED**	Action blocked by backend guard (if prev == Revoked return)	Preserved permanently as an immutable audit record

Comprehensive Role-Permission Reference Matrix

Application Capability	Security Guard	Warden	Campus Supervisor	Chief Security Officer (CSO)	Manager	Institute Manager	Admin
Initiate Face Scan at Gate	Yes	Yes (Hostel)	No	No	No	No	No
Manual Register No. Fallback	Yes	Yes	No	No	No	No	Yes
Log Unauthorized Incident Form	Yes	Yes	No	No	No	No	Yes
Receive Real-time Alert Badges	No	No	Yes	Yes	No	No	Yes
Authorise Pending Incident	No	No	Yes	Yes	No	No	No
Mark Unauthorised (w/ Action Taken)	No	No	Yes	Yes	No	No	No
Revoke Incident Decision	No	No	Yes	Yes	No	No	No
View Dedicated Action Taken Column	No	No	Yes	Yes	Yes	Yes	Yes
Export Entry Logs (CSV / PDF)	No	No	Yes	Yes	Yes	Yes	Yes
Operational Record Scope	Own Gate	Hostel Gate	Assigned Scope	Cross-Campus Scope	Read-Only	Read-Only	Full System
User & Role Provisioning	No	No	No	No	No	No	Yes